Applying XGBoost for Fault Prediction in Industrial Production Line

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Abstract:IntheeraofIndustry4.0,the levelofintelligenceand automationof productionlinesiscrucialforimproving productionefficiency.Thisstudyaddresses theissueoffaultpredictioninindustrial production lines by constructing an automaticalarmmodelusingXGBoostand neuralnetworktechnologytoenhancethe intelligence of production lines and optimize scheduling.Byanalyzingthecharacteristics offaultdataandusingcorrelationmatrices andtimeseriesdifferencingmethodsto build feature engineering, the model achievesaprecisionrateofupto97.99%, effectively predicting fault trends. Furthermore,themodelisappliedtoactual datatoautomaticallyalarmfaultsand statisticallyanalyzefaultfrequencyand duration. At the same time, by using correlationanalysisandmultiplelinear regression models, the study calculates production and qualification rates, revealing their relationships with production lines and operators, and presents them in graphical form. The modelsandmethodsinthisstudyhave practicalapplicationvalueforimproving industrial production efficiency.

Keywords:XGBoost;MachineLearning; Fault Prediction; Industrial Automation

1. Introduction

1.1 Research Background

Withtherapiddevelopmentofinformation technology,industrialproductionlinesare graduallytransitioningtowardsintelligence andautomation.Theapplicationofintelligent control technology enables automated productionlinestoautomaticallycomplete processessuchasitemconveyance,material filling, product packaging, and quality

inspection, greatly improving production efficiencyandproductqualitywhilereducing productioncosts.However,asthescaleand complexity of industrial production continue to expand,thechallengesfacedbyautomated productionlinesarebecomingincreasingly prominent. Traditional automated systems often lack sufficient intelligence and flexibility to effectively deal with changes and exceptions in the production process. Moreover, there is a lackofeffectiveinterconnectionbetweenthe equipmentandsystemsonthe production line, leadingtoinefficientinformationtransferand collaboration,which affectsoverall production efficiency andquality management. Therefore, how to further enhance the intelligence level of automatedproductionlinesandoptimizethe coordinationandflexibilityoftheproduction processhasbecomeakeyissuethatneedsto be addressed in the current industrial field [1].

1.2 Literature Review

Theintegrationofmachinelearning(ML) techniques with industrial processes, particularlyinthecontextoffaultprediction and maintenance, has seen significant advancementsinrecentyears.Thissection provides a comprehensive review of the literature pertaining to the application of ML in production lines and pipeline safety assessment, highlighting the key findings and methodologies employed.

The proliferation of Industry 4.0 has necessitated the adoptionof MLfor enhancing theefficiencyandreliabilityofproduction lines.Kangetal.[2]conductedasystematic literature review,identifying the application of ML in various industrial domains, with a focus onqualitycontrolandfaultdiagnosis.Their study underscored the dominance of supervisedlearningandthefrequentuseof artificialneural networks(ANN)inaddressing production line problems.

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The importance of data analytics in production lines has been emphasized by several researchers.Forinstance,Crespinoetal.[3] highlightedthechallengesinhandlingthe increasing data volumes in aerospace manufacturing, where real-time predictive analysis can improve output quality by identifying anomalies. Nakazawa and Kulkarni [4] utilized deep convolutional encoder-decoderneuralnetworksfordetectingwafer map defect anomalies in semiconductor manufacturing,demonstratingthepotentialof deep learning in defect detection.
Predictivemaintenance(PdM)hasemergedas acriticalareainindustrialassetmanagement. Paolantietal.[5] proposedamachinelearning approach for predictive maintenance in industry 4.0, emphasizing the need for condition monitoring and remaining useful life prediction.Yuetal.[6]presentedaglobal manufacturingbigdataecosystemforfault detection in predictive maintenance, showcasingthesynergybetweenbigdataand ML in enhancing maintenance strategies.
The application of ML in predictive maintenancehasbeenfurtherexploredby AydemirandPaynabar[7],whofocusedon image-basedprognosticsusingdeeplearning approaches. Similarly, Weber and Reimann[8] introducedaplatformtomanagemachine learningmodelsinIndustry4.0environments, highlightingthepracticalimplementationof ML in industrial settings.
Pipelinesafety,acriticalaspectoftheoiland gasindustry,hasalsoseentheapplicationof ML techniques. Elshaboury et al. [9] developed data-driven models for forecasting failure modes in oil and gas pipelines, using multilayerperceptron(MLP)neuralnetworks, radialbasisfunction(RBF)neuralnetworks, andmultinomiallogistic(MNL)regression. Theirmodelsachievedhighaccuracyrates, demonstratingtheeffectivenessofMLin predicting pipeline failures.
Liu et al. [10] proposed an XGBoost algorithm-basedmodelforthesafetyassessmentof

Pang [11]presentsadeeplearning-based approachforadaptivefaultpredictionand maintenanceinproductionlines,addressing limitationsintraditionalmethods.Thestudy introducesamodelthatincorporateswide convolutionalfeatureextraction,customized gating, and multi-layered progressive extractionmodules.ItutilizesWasserstein distanceforfaultstagedivisionandemploys L2regularizationandneurondropoutfor optimization,enhancingpredictionaccuracy andmaintenanceefficiency.Thisresearch contributestothefieldbyofferingamore precise and adaptable strategy for fault prediction,whichiscrucialforimproving productionlineperformanceandreducing operational costs.

1.3 Problem Formulation

Theresearchquestionsinthispapermainly focusonhowtoenhancetheintelligencelevel of automated production lines and optimize the coordinationandflexibilityoftheproduction process. Key issues include:

1)Howtobuildaneffectivefaultalarmmodel toachieveautomaticandimmediatefault alarms,reducingproductioninterruptionsand economic losses.

2)How to optimize personnel allocation throughdataanalysisandmodelconstruction to reduce resource waste and improve production efficiency.

3)How to address the lack of effective interconnection between equipment and systemsontheproductionlinetoimprove information transfer and collaboration efficiency.

Thispaperwillexploreanautomaticalarm model based on XGBoost and neural networks, andhowtoimprovetheintelligencelevelof productionlinesandoptimizetheproduction processthroughdataminingandmachine learningtechnology.Theultimategoalofthe researchistoprovideafeasiblemethodand technical route for the automatic recognition of production line faults and personnel allocation, inordertoimproveproductionefficiencyand product quality, and reduce production costs.

2. Materials and Methods

2.1 Dataset

The dataset used in this study covers 37 fields,

includingdate,time,productionlinenumber,

3. Analysis and Results

3.1 Data Processing

Beforeconstructingaproductionlinefault predictionmodel,itisessentialtoconductan in-depth analysis of the production line data to identifyandextractkeydatafeatures.This processinvolvesnotonlyfeatureengineering for fault data but also plays a crucial role in the **to** accuracy and generalization capability of subsequent models.

Firstly,datapreprocessingiscarriedout:(1) Datacleaning:Removeinvalidorincomplete records to ensure the consistency and reliability of the dataset. (2) Format standardization:Convertalldataintoaunified formatforeaseofsubsequentprocessingand analysis. (3) Missing value treatment: Interpolateordeletemissingdatatoprevent informationlossfrombiasingthemodel.(4) Outlierdetection:Identifyandhandleoutliers toreducetheiradverseeffectsonmodel training.

Next,featureengineeringisconductedto extractusefulinformationfromthedata:(1) Feature selection: Identify features most relevanttofaultprediction,reducingmodel complexity and improving predictive efficiency. (2) Feature transformation: Standardizeornormalizefeaturestothesame scale, enhancing the model's convergence speedandaccuracy.(3)Featureconstruction: Createnewfeaturesbasedonexistingdatato reveal hidden patterns and relationships.

Finally, statistical analysis methods are sued

explore thecorrelationsbetween various faults and other data features.

From thecorrelationmatrixheatmapshownin Figure 1, it is evident that there is a significant correlationbetweenmostdatafeaturesand faultdata.Thisindicatesthatfaultsareoften directlycausedbytheoperationalstatusof earlierprocesses,suchasthenumberoftimes the capping device presses down on bottle caps ontoproductbottlesandthenumberoftimes thescrewingdevicescrewsontheproduct bottlecaps,whichhaveadirectimpactonthe occurrenceofsubsequentfaults.Throughthis intuitivevisualizationmethod,wecanidentify keyprocessdatathatarecloselyrelatedtothe occurrence of faults, providing important input features for subsequent fault prediction models.

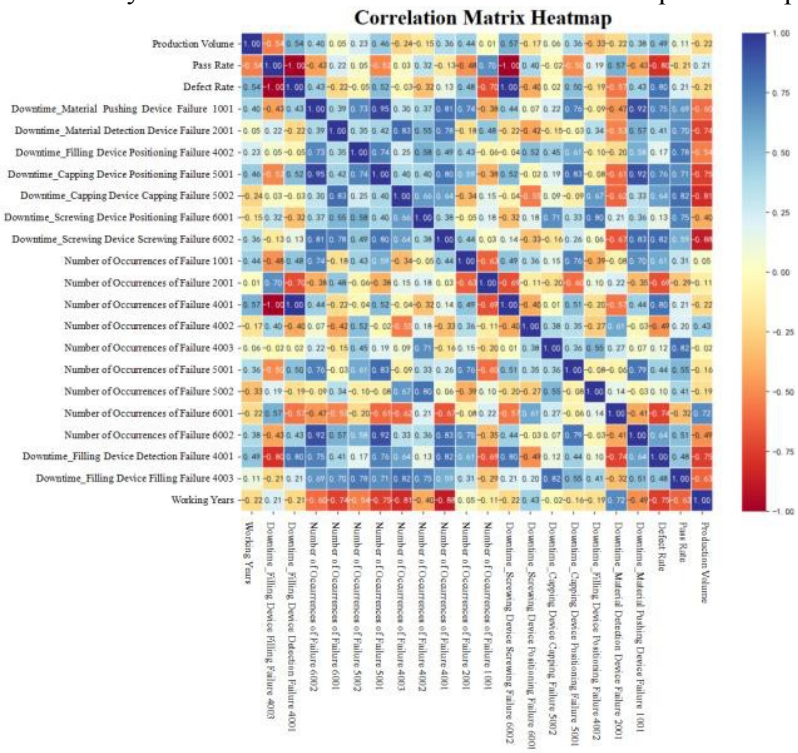

Figure 1. Correlation Matrix Heatmap

3.2 Model Development

preprocessed using the pandas library in Python.Thespecificstepsinclude:handling

ThisstudyemploystheXGBoostalgorithmto construct a **fualt** prediction model for production lines. Initially, the dataset is missingvaluesinthefaultdatabyiterating through the data to mark the start time of faults and calculating the duration of faults, filling all

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missing values with 0. Additionally, to address the issue of data imbalance, this study employs oversamplingmethodstobalancetheclass distribution. During the feature extraction phase,theprocesseddatasetisdividedintoa featuresetandalabelset,whicharethen converted into tensor format under the PyTorchframeworkforsubsequentmodel training.

```
from imblearn.over_sampling import SMOTE
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from sklearn.model_selection import train_test
_split
from torch.utils.data import DataLoader, Tenso
rDataset
from imblearn.over_sampling import SMOTE
import xgboost as xgb
from sklearn.metrics import accuracy_score from
sklearn.utils import resample
from tqdm import tqdm
import os

def incremental_xgboost_training (file_path, e
xisting_model, new_model_name, params=No
ne, num_rounds=100):
# Load data
df = pd.read_csv(file_path)

    # Data preprocessing
    df.iloc[:, 2] = df.iloc[:, 2].str.extract('(\d+)').
astype(int)
df.iloc[:, -9:] = df.iloc[:, -9:].applymap(lamb da
x: 1 if x != 0 else 0)
df = df.astype(int)

 error_df = pd.DataFrame()
 fault_columns = [col for col in df.columns if
 'fault ' in col]

# Calculate fault duration
for col in fault_columns:
        error_df[f'{col}_start'] = (df[col]! = 0) &
(df[col]. shift (1) == 0)
        error_df[f'{col}_duration']= df.groupby
((df[col] == 0).cumsum())[col].transform('coun
t') * error_df[f'{col}_start']
        duration_temp = pd.Series(index=error_d
```

```
for i in error_df.index[error_df[f'{col}_sta rt']]:
        duration_temp.iloc[i:i + error_df.at[i, f'
{col}_duration']] = error_df.at[i, f'{col}_durati
on']

error_df[f'{col}_duration']= duration_te
mp.fillna(0).astype(int)

    # Add fault duration to data
    for col in fault_columns:
    df[f'{col}_duration'] = error_df[f'{col}_d
uration']

    X = df.iloc[:, :-18].values y =
    df.iloc[:, -18:-9].values

    X_tensor = torch.tensor(X, dtype=torch.float
32)
    y_tensor = torch.tensor(y, dtype=torch.float
32)

    dtrain = xgb.DMatrix(X_tensor, label=y_ten
sor)

    # If parameters are not provided, set default
parameters
    if params is None:
    params = {
        'max_depth': 3,
        'eta': 0.3,
        'objective': 'binary:logistic',
        'eval_metric': 'auc',
        'num_feature': 28 # Set the number of f
eatures to 28
    }

    # Continue training the model
    model = xgb.train(params, dtrain, num_roun
ds, xgb_model=existing_model)

    # Rename the model
    model.set_attr (name='name', value=new_m
odel_name)

    return model
```

Subsequently,thepreprocessedfeaturesets andlabelsetsareconvertedintotheDMatrix data structure required by the XGBoost algorithm. Building on this, this study employs anincrementallearningstrategy,combining existingmodelswithnewdatafortraining.In

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keyparameters,includingthemaximumdepth of the trees and the learning rate. After making predictionsonthetestset,thepredicted probabilityvaluesareconvertedintointeger form prediction labels, and the model's accuracy,precision,recall,andF1scoreare calculatedaftermultiplerounds oftrainingto comprehensively evaluate the model's performance. Through the aforementioned steps,thisstudysuccessfullyconstructedan XGBoostfaultpredictionmodel,providing technical support for the automatic alarming of production line faults.

Table 1. Model Training Results

Evaluation Metrics	Values
Accuracy	97.9745%
Precision	0.0545
Recall	0.16565
F1 Score	0.08204

Table1presentstheperformanceevaluation results of the model. The accuracy rate reaches 97.9745%, indicating that the model has a high overall predictive accuracy. However, the precisionisonly0.0545,whichmaybe attributedtothehighsensitivityofthemodel leadingtomorefalsepositives.Therecallrate is 0.16565, indicating that a significant number of positive samples are not correctly identified, suggestingadeficiencyinthemodel'sability torecognizepositivesamples.TheF1scoreis 0.08204,furtherreflectingtheimbalancein modelperformance.Accuracyreflectsthe overallpredictiveaccuracyofthemodel, precisionfocusesontheaccuracyofpositive predictions,recallemphasizestheabilityto identify positive samples, and the F1 score is a comprehensivereflectionofprecisionand recall.Byutilizingspecificmachinelearning librariestoevaluatemodelpredictions,a deeper understanding of the model's performance can be gained, providing important reference for subsequent model optimization and practical application.

3.3 Prediction

To further understand and address the issues of faultswithinproductionlines,andtoenhance

robustnessofpredictivemodels,capableof elucidating the probabilistic distributional characteristics of the model's forecasts.

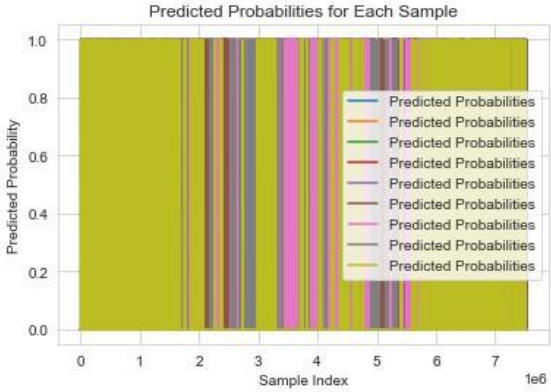

Figure 2. Predicted Probabilities for Each Sample

As depicted in Figure 2, the histogram demonstratesthatthepredictedprobabilities are primarily distributed within the intervals of 0-0.2and0.6-0.8.Withinthe0-0.2range, thedatapointsarerelativelyconcentrated, indicatingthatthemodelpredictsahigher number of negative class samples (i.e., no fault occurrence) with a consistent probability, reflectingahighercertaintyinthemodel's predictions for this range. Conversely, the 0.6 - 0.8rangealsocontainsacertainnumberof data points, representing the model's prediction of positive class samples (i.e., fault occurrence),buttheprobabilitydistributionis more dispersed, suggesting a lower certainty in themodel's predictions forthissegment. Overall, the distribution of the model's predictive outcomes exhibits a certain skewness, which may be related to the frequencyoffaultoccurrencesintheactual data.Byanalyzingthehistogram,wecangain a visual understanding of the model's performance tendencies across different predictioncategories,providingabasisfor subsequentmodeloptimization.Forinstance, targetingintervalswithambiguousprediction probabilities,furtheradjustmentstomodel parameters or improvements in feature engineeringcouldbeimplementedtoenhance themodel'spredictiveaccuracyandstability. Additionally,calculatingthestartandend positionsoffaults,aswellastheirduration,is crucialforunderstandingtheoccurrenceand progression of faults, offering important referencesforfaultdiagnosisandrepair.By creatingaDataFrametorecordthestarttimes and durations of faultsand merging it with the

original data,wecan obtain richer information for in-depth analysis and processing. Renamingthecolumnheadersandresetting the index enhances the readability and usabilityofthedata,facilitatingsubsequent data analysis and processing. Saving the resultingDataFrameasaCSVfilefacilitates integrationandsharingwithothertoolsand systemsforfurtheranalysisandprocessing. Overall,theaforementionedprocessesand outcomesareofsignificantimportancefor thoroughlyunderstandingandaddressingfault issuesinproductionlines,andforimproving production efficiency and quality. The prediction results are presented in Table 2.

Table 2. Model Prediction Results (Partial Data Shown)

Fault Id	1001			1002			...
S/N	dt.	Start time	Dur. (sec.)	dt.	Start time	Dur. (sec.)	...
0	12	9072	6	4	8258	3	...
1	15	5250	157	4	8435	4	...
2	17	20882	9	4	10742	7	...
3	17	20902	140	4	10939	7	...
4	24	25432	9	5	10939	7	

4. Conclusions

Thisstudyhasdemonstratedthatthefault prediction model based on the XGBoost algorithmishighlyeffectiveandpracticalin real-world applications. The model's high accuracy rate has proven its predictive capabilities in complex industrial environments, particularly in reducing production interruptions and optimizing resourceallocation.Despitethechallenges facedinprecisionandrecallrates,these metricsalsorevealareasforimprovementin themodel'sidentificationofspecificfault types.Futureworkwillfocusonenhancing these performance indicators and extending the modeltomoreproductionlinescenariosto verifyitsgeneralizationcapabilities.Overall, thisstudynotonlyprovidesaneffectivetool for fault prediction in production lines but also offers strong technical support for the intelligentandautomatedproductioninthe context of Industry 4.0.

4.1 Model Advantages

The XGBoost model proposed in this study has shown significant advantages in fault prediction for production lines, primarily reflectedinitsexceptionalaccuracyrateofup to97.99%.Thisnotonlyprovesthemodel's reliability in predicting faults in most cases but increasingresponsespeedandefficiency,and aidingintheredutionofresourcewasteand enhancementofoverallproductionefficiency, thusstrengtheningthescientific basisfor decision-making.

4.2 Model Disadvantages

Althoughthemodelexcelsinaccuracy,its performanceinprecisionandrecallrates leavesroomforimprovement,whichmay pointtoissueswiththemodel'sperformance under specific conditions, especially in identifyingpositivesampletypes.Thiscould beduetodataimbalanceorthemodel's insufficient sensitivity to certain features. Therefore, further model needed to enhance its sensitivity and

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industrialsectors,thereispotentialtoimprove theoverallefficiencyandqualityofindustrial production.

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